



Problem Statement #1: AI-Based Human Behaviour Monitoring Using Pose Estimation and Gesture Analysis

While conducting an examination, continuous manual observation is difficult, leading to situations where suspicious behaviours may go unnoticed or be detected too late. The challenge is to develop an AI-powered video analytics system capable of monitoring seated students in real time using pose estimation and gesture analysis techniques. The system should analyse body posture, head orientation, and hand movements from surveillance camera feeds to identify behaviours that may indicate potential examination misconduct, such as repeated sideward glances, excessive head turning, body rotation toward neighbouring students, unusual hand movements, or attempts to communicate non-verbally. Unlike traditional surveillance systems that rely heavily on facial recognition or manual review of recorded footage, the proposed solution should focus on skeletal keypoints and motion patterns to infer behavioral anomalies while preserving student privacy. The system must operate in real time, track multiple students simultaneously, and generate explainable alerts that highlight specific observed behaviors rather than making direct accusations of cheating.

The system should be robust to variations in seating arrangements, camera angles, lighting conditions, and student demographics while maintaining high accuracy and minimizing false alarms.

Objectives

- Detect and track multiple seated students from live video feeds.
- Extract human pose landmarks and movement patterns.
- Identify suspicious head, body, and hand movements associated with potential examination misconduct.
- Identify objects like Mobile Phone, Paper/Chit or any unwanted objects as may be feasible
- Generate real-time alerts with interpretable explanations.
- Provide behavioral analytics and event logs for invigilator review.
- Ensure privacy-conscious monitoring by avoiding identity-based recognition wherever possible.

Expected Outcome A scalable AI-driven examination monitoring system that assists invigilators in maintaining examination integrity by automatically identifying and reporting unusual behavioral patterns in examination halls through pose estimation and gesture detection.

Risk Factor	Description	Impact
False Positives	Normal student actions (thinking, stretching, looking around) may be incorrectly flagged	Reduced trust in the system and unnecessary invigilator intervention.
False Negatives	Actual suspicious behavior may not be detected.	Potential examination malpractice goes unnoticed.
Occlusion	Students blocking each other from the camera view.	Incomplete pose estimation and tracking errors.
Poor Lighting Conditions	Low light, shadows, or glare affecting video quality.	Reduced detection accuracy.
Camera Blind Spots	Areas not covered by surveillance cameras.	Students in blind spots cannot be monitored effectively.
Head Pose Estimation Errors	Difficulty accurately detecting viewing direction from distant cameras.	Incorrect behaviour classification.
Multi-Person Tracking Failure	Identity switching or loss of student tracking across frames.	Incorrect attribution of suspicious activities.
Network Failure	Loss of connectivity between cameras and processing servers.	Interrupted monitoring and missed events.
Hardware Failure	Camera, GPU, storage, or server malfunction during examinations.	System downtime and loss of monitoring capability.
Privacy Concerns	Student concerns regarding surveillance and data collection.	Regulatory, ethical, and acceptance challenges.

Technology Challenges:

1. Differentiating Suspicious and Natural Behavior

Students may naturally:

- Look around due to distractions.
- Stretch or adjust their posture.
- Raise their heads while thinking.
- Turn briefly toward invigilators or classroom boards.

The system must distinguish between normal examination behavior and genuinely suspicious patterns to avoid excessive false alerts.

2. Camera Placement and Viewing Angles

Examination halls vary significantly in:

- Seating layouts.
- Camera heights.



- Camera orientations.
- Student density.

Poor camera placement can lead to incomplete body visibility, making pose estimation less reliable.

3. Occlusion and Crowded Environments

Students may partially block one another, causing:

- Missing body landmarks.
- Inaccurate gesture detection.
- Tracking failures during movement.

Maintaining accurate monitoring in densely populated halls is a major challenge.

4. Lighting and Video Quality Variations

Real-world surveillance footage may suffer from:

- Low lighting.
- Shadows.
- Glare from windows.
- Low-resolution cameras.
- Motion blur.

The system must remain reliable across diverse environmental conditions.

5. Multi-Person Tracking Consistency

The system must continuously track individual students throughout the examination without:

- Identity switching.
- Track loss.
- Duplicate detections.

Reliable long-duration tracking is critical for behavior analysis.

6. False Positives and Alert Fatigue

Frequent incorrect alerts can:

- Distract invigilators.
- Reduce trust in the system.
- Increase monitoring workload.

A deployment-ready solution must prioritize precision and explainable decision-making.

7. Behavioral Diversity

Student behavior varies across:

- Age groups.
- Examination formats.
- Cultural contexts.



- Individual habits.

Models trained on limited datasets may not generalize effectively to different examination environments.

8. Real-Time Processing at Scale

Large examination centers may contain:

- Hundreds of students.
- Multiple camera streams.
- Long examination durations.

The system must deliver low-latency performance while managing computational resources efficiently.

9. Privacy and Ethical Considerations

Institutions may have concerns regarding:

- Student surveillance.
- Data storage policies.
- Consent requirements.
- Compliance with privacy regulations.

The system should prioritize anonymized behavioral analysis rather than identity recognition.

10. Explainability and Human Oversight

An alert should never be treated as definitive evidence of cheating.

The system must provide:

- Visual evidence.
- Behavioral explanations.
- Confidence scores.
- Event timelines.

Final decisions should remain with human invigilators and examination authorities.

11. Dataset Collection and Annotation

Developing robust models requires:

- Diverse examination hall videos.
- Accurate behavior annotations.
- Examples of both normal and suspicious activities.

Obtaining such datasets while respecting privacy and ethical standards can be difficult.

12. Generalization Across Institutions

A model performing well in one institution may fail in another due to differences in:

- Camera infrastructure.
- Seating arrangements.



- Student behavior patterns.
- Examination procedures.

The solution should be adaptable and configurable for deployment across multiple educational environments.

Hackathon Extension Goal: Beyond simple detection, the system should produce a **behavioral risk score and explainable event summaries**, enabling invigilators to focus attention on students exhibiting sustained patterns of unusual behavior rather than isolated movements. This improves practicality, fairness, and trustworthiness in real-world deployment.

Component	Specification	Purpose
Edge AI Device (Optional)	NVIDIA Jetson Orin Nano or NVIDIA Jetson AGX Orin	Local real-time pose estimation and inference
AI Processing Server	16–32 Core CPU, 64–128 GB RAM	Centralized video analytics and behavior monitoring
GPU	NVIDIA RTX 4070 / NVIDIA RTX 4080 / NVIDIA RTX 4090	Accelerate pose estimation and gesture detection models
Storage	2–8 TB NVMe SSD	Store video recordings, logs, and analytics data
Backup Storage Server (Optional)	NAS or Dedicated Storage Server	Long-term storage of examination footage and event logs

Deployment Size	Students	Cameras Required	Processing Hardware
Small Hall	50–100	2–4 Cameras	NVIDIA Jetson AGX Orin
Medium Hall	100–300	4–8 Cameras	AI Server + NVIDIA RTX 4080
Large Hall	300–1000	8–20 Cameras	Multi-GPU AI Server
Exam Center	1000+	20–30+ Cameras	GPU Cluster + Central Monitoring System



Problem Statement #2: Offline Video Segmentation and ROI Detection Using Motion Estimation

Examination authorities often need to review large volumes of recorded surveillance footage after examinations to investigate incidents, verify suspicious activities, and analyze student behavior. Manual review of hours of video footage is time-consuming, labor-intensive, and prone to human error, making it difficult to efficiently identify relevant events.

The challenge is to develop an AI-powered offline video analytics system that automatically segments recorded examination hall videos and identifies Regions of Interest (ROI) using motion estimation techniques. The system should analyze temporal motion patterns within video recordings to detect areas where significant activity occurs and segment the footage into meaningful events for further review.

Unlike traditional video review methods that require continuous manual observation, the proposed solution should automatically identify motion-based regions, filter out static scenes, highlight periods of unusual activity, and generate summarized outputs for investigators and examination authorities. The system should operate on previously recorded videos, enabling efficient post-examination analysis without requiring real-time processing.

The solution should provide event-based video segmentation, ROI visualization, motion heatmaps, activity timelines, and searchable event logs to reduce the effort required for manual footage examination.

Objectives

- Process recorded examination hall videos offline.
- Detect motion regions using frame-to-frame motion estimation techniques.
- Automatically identify Regions of Interest (ROI) containing significant activity.
- Identify external objects like Mobile Phone, Paper/Chits or another other prohibited object as may be feasible
- Segment long video recordings into activity-based clips.
- Generate motion heatmaps and activity timelines.
- Reduce manual video review effort by highlighting relevant events.
- Provide event summaries and visual analytics for investigators.

Expected Outcome

A scalable offline video analytics system capable of automatically segmenting examination hall recordings and identifying motion-based Regions of Interest, enabling faster and more efficient post-examination investigations.

Challenges for Real-World Deployment

Challenge	Description
Crowded Examination Halls	Simultaneous movements from multiple students can create overlapping motion regions.
Subtle Motion Detection	Small movements such as head turns, hand gestures, or object exchanges may be difficult to distinguish from normal activity.
Camera Noise	Video compression artifacts, sensor noise, and recording distortions may generate false motion detections.
Lighting Variations	Changes in illumination, shadows, and reflections can affect motion estimation accuracy.
Camera Vibration	Slight camera movement can produce false motion across the entire frame.
Long Video Duration	Examination recordings may span several hours, requiring efficient processing and storage management.
ROI Boundary Accuracy	Accurately localizing the area responsible for motion can be challenging in dense environments.
Background Motion	Fans, curtains, lighting fluctuations, and environmental changes may create unwanted motion signals.
Event Segmentation Quality	Determining the start and end of meaningful events requires robust temporal analysis.
Multi-Camera Synchronization	Combining ROI information from multiple camera feeds may require timestamp alignment.
Scalability	Processing large numbers of recorded videos across multiple examination centers can be computationally intensive.
False Event Generation	Minor or irrelevant movements may be incorrectly classified as important events.

Hardware Requirements for Final Deployment

Component	Specification	Purpose
GPU	NVIDIA RTX 4070 / NVIDIA RTX 4080 / NVIDIA RTX 4090	Accelerated video processing and segmentation
RAM	64–128 GB	Handling large video datasets
Storage	4–20 TB SSD/HDD	Archival and processed video storage
Backup Storage	NAS or Storage Server	Long-term retention of recordings and analytics
Monitoring Workstation	16–32 GB RAM, Dual Monitors	Reviewing segmented events and ROI outputs
Network Infrastructure	Gigabit Ethernet	Transfer of recorded footage from surveillance systems

Risk Factors

Risk Factor	Description	Impact
False Motion Detection	Noise, shadows, or lighting changes interpreted as movement.	Increased number of irrelevant event segments.
Missed ROI Detection	Small or subtle movements not detected.	Important events may be overlooked.
Over-Segmentation	Excessive splitting of video into many small segments.	Reduced usability and increased review effort.
Under-Segmentation	Multiple events merged into a single segment.	Difficult event analysis.
Camera Motion Effects	Vibration or repositioning causing false motion regions.	Inaccurate ROI generation.
Storage Constraints	Large examination archives require substantial storage resources.	Increased infrastructure costs.
Processing Delays	Long analysis times for high-resolution videos.	Delayed investigation results.
ROI Localization Errors	Incorrect identification of activity regions.	Misleading event summaries.
Environmental Variability	Different hall layouts and camera configurations affect performance.	Reduced generalization.
Data Security Risks	Unauthorized access to recorded examination footage.	Privacy and compliance concerns.
Hardware Failures	Storage or server failures during analysis.	Data loss and processing interruptions.
Scalability Challenges	Increasing video volume across institutions.	Resource bottlenecks.

Mitigation Strategy

The system should function as an **offline investigation support tool** that automatically prioritizes and summarizes recorded footage for human reviewers. Motion-based ROIs and segmented events should be accompanied by timestamps, visual overlays, activity scores, and confidence metrics, allowing investigators to quickly focus on potentially relevant portions of examination recordings while maintaining human oversight in the final review process.